

Thermal Stress and Recovery of Reef-Building Corals: A Thesis Summary

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Abstract

This thesis examines how marine heatwaves trigger the breakdown of the symbiosis between reef-building corals and their photosynthetic algae (zooxanthellae, family Symbiodiniaceae). When sea surface temperatures stay about 1 degree C above the summer maximum for several weeks, corals expel their algae and turn white. Without the sugars supplied by the algae, the coral starves unless temperatures return to normal quickly.

Methods

We monitored 42 permanent transects on three reefs between 2019 and 2024. Heat stress was measured in degree heating weeks (DHW) derived from satellite temperature data. Colonies were tagged and photographed every two months, and tissue samples were genotyped to identify the dominant symbiont clade.

Results

Bleaching began at 4 DHW and mortality rose sharply above 8 DHW. Branching *Acropora* species suffered up to 70 percent mortality, while massive *Porites* colonies mostly recovered within six months. Colonies hosting heat-tolerant *Durussdinium* symbionts bleached less, but grew about 30 percent slower in normal years, suggesting a trade-off. Reefs with high herbivorous fish biomass showed faster recovery because grazing kept seaweed from overgrowing dead coral skeletons.

Discussion

Recurrent heatwaves now arrive every 3 to 6 years, faster than the 10 to 15 years that reefs need to recover. Local management - reducing nutrient runoff and protecting grazing fish - can buy time, but cannot replace cutting global greenhouse gas emissions.

Conclusion

Coral communities are shifting toward slower-growing, stress-tolerant species, with less structural complexity and less habitat for reef fish.